

Solution-Chemistry Dictionary

ID	Class	Canonical name	Synonyms (JSON)	Parameters (JSON)	Version	Created
578	buffer	ADA	["N-(2-acetamido)iminodiacetic acid"]	{"pKa_25C": 6.6, "useful_pH_range": [6.0, 7.2]}	1	2026-05-20 22:24:11.306852+01
571	buffer	Bicine	["N,N-bis(2-hydroxyethyl)glycine"]	{"dpKa_dT": -0.018, "pKa_25C": 8.35, "useful_pH_range": [7.6, 9.0]}	1	2026-05-20 22:24:11.306852+01
572	buffer	Bis-Tris	["bis(2-hydroxyethyl)amino-tris(hydroxymethyl)methane", "bistris", "bis tris", "BisTris", "deuterated bis-tris", "d-Bis-Tris", "d-BisTris", "Bis-Tris buffer"]	{"pKa": [{"source": "Goldberg 2002", "step": 1, "value": 6.46}], "pKa_25C": 6.46, "source": "Goldberg-Kishore-Lennen 2002", "species": [{"charge": 1, "label": "Bis-Tris-H+", "step": null}, {"charge": 0, "label": "Bis-Tris", "step": 1}], "useful_pH_range": [5.8, 7.2]}	1	2026-05-20 22:24:11.306852+01
585	buffer	Bis-Tris-Propane	["BisTrisPropane", "Bis-Tris Propane", "BTP", "1,3-bis(tris(hydroxymethyl)methylamino)propane"]	{"dpK1_dT": -0.0085, "notes": "Diprotic buffer; spans pH 6.3-9.7 with two non-overlapping pKa regions. Distinct from Bis-Tris (single pKa 6.46).", "pK1_25C": 6.8, "pK2_25C": 9.0, "source": "Goldberg 2002", "species": [{"charge": 2, "label": "BisTrisPropane-H2^2+", "step": null}, {"charge": 1, "label": "BisTrisPropane-H+", "step": 1}, {"charge": 0, "label": "BisTrisPropane", "step": 2}], "useful_pH_range_pK1": [6.3, 7.3], "useful_pH_range_pK2": [8.3, 9.7]}	1	2026-05-20 22:24:11.306852+01
566	buffer	HEPES	["Na HEPES", "K HEPES", "Na-HEPES", "K-HEPES", "HEPES-NaOH", "HEPES-KOH", "4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid", "HEPES buffer"]	{"dpKa_dT": -0.014, "pKa_25C": 7.55, "species": [{"charge": 0, "label": "HEPES-H", "step": null}, {"charge": -1, "label": "HEPES-", "step": 1}], "useful_pH_range": [6.8, 8.2]}	1	2026-05-20 22:24:11.306852+01
576	buffer	HEPPS	["EPPS", "4-(2-hydroxyethyl)piperazine-1-propanesulfonic acid"]	{"pKa_25C": 8.0, "useful_pH_range": [7.3, 8.7]}	1	2026-05-20 22:24:11.306852+01
567	buffer	MES	["Na MES", "MES-NaOH", "2-(N-morpholino)ethanesulfonic acid", "mes buffer"]	{"dpKa_dT": -0.011, "pKa": [{"source": "Goldberg 2002", "step": 1, "value": 6.1}], "pKa_25C": 6.1, "source": "Goldberg-Kishore-Lennen 2002", "species": [{"charge": 1, "label": "MES-H+", "step": null}, {"charge": 0, "label": "MES", "step": 1}], "useful_pH_range": [5.5, 6.7]}	1	2026-05-20 22:24:11.306852+01
568	buffer	MOPS	["Na MOPS", "MOPS-NaOH", "3-(N-morpholino)propanesulfonic acid"]	{"dpKa_dT": -0.011, "pKa_25C": 7.2, "species": [{"charge": 0, "label": "MOPS-H", "step": null}, {"charge": -1, "label": "MOPS-", "step": 1}], "useful_pH_range": [6.5, 7.9]}	1	2026-05-20 22:24:11.306852+01
569	buffer	PIPES	["Na PIPES", "piperazine-N,N'-bis(2-ethanesulfonic acid)", "PIPES buffer"]	{"dpKa_dT": -0.0085, "pKa_25C": 6.76, "species": [{"charge": -1, "label": "HPIPES-", "step": null}, {"charge": -2, "label": "PIPES^2-", "step": 1}], "species_note": "piperazine disulfonate; pKa2 step is monoanion->dianion", "useful_pH_range": [6.1, 7.5]}	1	2026-05-20 22:24:11.306852+01
570	buffer	TAPS	["[(2-hydroxy-1,1-bis(hydroxymethyl)ethyl)amino]-1-propanesulfonic acid"]	{"dpKa_dT": -0.018, "pKa_25C": 8.4, "useful_pH_range": [7.7, 9.1]}	1	2026-05-20 22:24:11.306852+01
577	buffer	TES	["N-tris(hydroxymethyl)methyl-2-aminoethanesulfonic acid"]	{"dpKa_dT": -0.02, "pKa_25C": 7.4, "useful_pH_range": [6.8, 8.2]}	1	2026-05-20 22:24:11.306852+01
575	buffer	Tricine	["N-(2-hydroxy-1,1-bis(hydroxymethyl)ethyl)glycine"]	{"dpKa_dT": -0.021, "pKa_25C": 8.05, "useful_pH_range": [7.4, 8.8]}	1	2026-05-20 22:24:11.306852+01
565	buffer	Tris	["Tris-HCl", "Tris-HCL", "Tris HCl", "tris-hydrochloride", "Trizma", "Trizma HCl", "Trizma base", "THAM", "tris base", "tris/HCl", "TRIS", "tris(hydroxymethyl)aminomethane", "d-Tris", "d-Tris-HCl", "d-Tris HCl", "d-Tris-HCl (pH 7.0)", "d-Tris-HCl(pH 7.0)", "d11-Tris", "deuterated Tris", "perdeuterated Tris", "tris-d11", "dtrishcl", "d-trishcl", "d-tris-hcl2", "d-trishcl(pH7.0)", "Tris buffer", "TRIS-maleate", "Tris-HCl(7.0)", "Tris-HCl(pH 7.0)", "Tris-d11-HCl"]	{"dpKa_dT": -0.028, "notes": "Strong T-dependence: room-T vs body-T differ by ~0.4 pH units.", "pKa": [{"source": "Goldberg 2002", "step": 1, "value": 8.06}], "pKa_25C": 8.06, "source": "Goldberg-Kishore-Lennen 2002", "species": [{"charge": 1, "label": "TrisH+", "step": null}, {"charge": 0, "label": "Tris", "step": 1}], "useful_pH_range": [7.0, 9.0]}	1	2026-05-20 22:24:11.306852+01
581	buffer	acetate	["Na acetate", "sodium acetate", "K acetate", "NH4 acetate", "ammonium acetate", "acetic acid", "HOAc/OAc-", "acetate buffer", "deuterated sodium acetate", "deuterated sodium acetate buffer", "na-acetate", "naac", "ch3coona", "d-NaOAc", "AcONa", "Acetate buffer Na", "potassium acetate"]	{"dpKa_dT": -0.0002, "pKa": [{"source": "CRC Handbook", "step": 1, "value": 4.76}], "pKa_25C": 4.76, "source": "CRC Handbook of Chemistry and Physics", "species": [{"charge": 0, "label": "AcOH", "step": null}, {"charge": -1, "label": "AcO-", "step": 1}], "useful_pH_range": [3.6, 5.6]}	1	2026-05-20 22:24:11.306852+01
583	buffer	borate	["sodium borate", "NaB(OH)4", "boric acid", "Na borate"]	{"dpKa_dT": -0.008, "pKa_25C": 9.24, "useful_pH_range": [8.5, 10.2]}	1	2026-05-20 22:24:11.306852+01
574	buffer	cacodylate	["Na cacodylate", "sodium cacodylate", "dimethylarsinic acid"]	{"notes": "Contains arsenic; uncommon in modern depositions.", "pKa_25C": 6.27, "useful_pH_range": [5.0, 7.4]}	1	2026-05-20 22:24:11.306852+01
584	buffer	carbonate	["Na carbonate", "K carbonate", "NaHCO3 / Na2CO3 mix", "bicarbonate", "Na bicarbonate"]	{"pK1_25C": 6.35, "pK2_25C": 10.33, "useful_pH_range_pK1": [5.5, 7.0], "useful_pH_range_pK2": [9.0, 10.7]}	1	2026-05-20 22:24:11.306852+01
582	buffer	citrate	["Na citrate", "sodium citrate", "K citrate", "citric acid", "ammonium citrate"]	{"dpK3_dT": -0.004, "notes": "Tri-protic; weak chelator (can bind Mg2+, Ca2+, Fe3+).", "pK1_25C": 3.13, "pK2_25C": 4.76, "pK3_25C": 6.4, "species": [{"charge": 0, "label": "H3Cit", "step": null}, {"charge": -1, "label": "H2Cit-", "step": 1}, {"charge": -2, "label": "HCit^2-", "step": 2}, {"charge": -3, "label": "Cit^3-", "step": 3}], "useful_pH_range_pK1": [2.5, 3.5], "useful_pH_range_pK2": [3.5, 5.5], "useful_pH_range_pK3": [5.5, 6.5]}	1	2026-05-20 22:24:11.306852+01
579	buffer	glycine	["Gly", "glycine buffer"]	{"dpK2_dT": -0.025, "notes": "Can also appear as the free amino acid (sample component) at high pH.", "pK1_25C": 2.34, "pK2_25C": 9.6, "useful_pH_range_pK1": [2.2, 3.6], "useful_pH_range_pK2": [8.6, 10.6]}	1	2026-05-20 22:24:11.306852+01
580	buffer	glycyl-glycine	["Gly-Gly", "glycylglycine"]	{"dpKa_dT": -0.025, "pKa_25C": 8.4, "useful_pH_range": [7.5, 8.9]}	1	2026-05-20 22:24:11.306852+01
573	buffer	imidazole	["imid", "imidazole buffer", "1,3-diazole"]	{"dpKa_dT": -0.02, "notes": "Coordinates many metals; in protein samples often present as a competitor for affinity-tag elution rather than as a buffer.", "pKa_25C": 7.0, "useful_pH_range": [6.2, 7.8]}	1	2026-05-20 22:24:11.306852+01
586	buffer	maleic acid	["maleate", "sodium maleate", "Na maleate", "maleate buffer"]	{"notes": "Diprotic; used in pH 5-6.5 range (around pK2). Sometimes combined with Tris (Tris-maleate buffer).", "pK1_25C": 1.92, "pK2_25C": 6.07, "source": "CRC Handbook", "species": [{"charge": 0, "label": "H2Mal", "step": null}, {"charge": -1, "label": "HMal-", "step": 1}, {"charge": -2, "label": "Mal^2-", "step": 2}], "useful_pH_range": [5.0, 6.5]}	1	2026-05-20 22:24:11.306852+01

ID	Class	Canonical name	Synonyms (JSON)	Parameters (JSON)	Version	Created
564	buffer	phosphate	["sodium phosphate", "Na phosphate", "K phosphate", "potassium phosphate", "phosphate buffer", "phosphate buffer saline (no NaCl)", "potassium phosphate buffer", "sodium phosphate buffer", "k-phosphate buffer", "nah2po4 buffer", "KP", "K-Pi", "Pi", "NaPi", "PINA", "PiNa", "PiNa", "PiNa buffer", "phosphate buffer Na", "k-po", "NaPO4 buffer", "NaH2PO4/Na2HP04 mix", "NaH2PO4-Na2HP04", "k2hpo4/kh2po4", "PBS", "PBS buffer", "na2po4", "kpi", "kpo4", "na phosphate", "napo4", "nahpo4", "po4", "kh2po4", "potasium phosphate", "na-phosphate", "phophate", "phospate", "posphate", "phospate buffer", "phohphate buffer", "Na phohphate buffer", "photasuim phosphate", "potassium phosphate buffer", "sodium posphate", "sodium phospate", "phosphate_buffer", "sodium_phosphate", "phosphate buffer K", "PHOSPHATE BUFFER K", "NAPHOS", "Na2HP04/NaH2PO4", "NaP"]	{"dpK2_dT": -0.0028, "notes": "pK2 is the buffering pK at biological pH. PBS = phosphate-buffered saline includes NaCl/KCl.", "pk1_25C": 2.15, "pk2_25C": 7.21, "pk3_25C": 12.4, "pKa": [{"source": "CRC Handbook", "step": 1, "value": 2.15}, {"source": "Goldberg 2002", "step": 2, "value": 7.21}, {"source": "CRC Handbook", "step": 3, "value": 12.4}], "source": "Goldberg-Kishore-Lennen 2002 (pK2, dpK2/dT); CRC Handbook (pK1, pK3)", "species": [{"charge": 0, "label": "H3PO4", "step": null}, {"charge": -1, "label": "H2PO4-", "step": 1}, {"charge": -2, "label": "HP04^2-", "step": 2}, {"charge": -3, "label": "P04^3-", "step": 3}], "useful_pH_range": [5.8, 8.0]}	1	2026-05-20 22:24:11.306852+01
596	chelator	BAPTA	["1,2-bis(o-aminophenoxy)ethane-N,N,N',N'-tetraacetic acid", "BAPTA-AM"]	{"notes": "Aromatic Ca2+ chelator; not present in our 656 working-set corpus as of 2026-05-20 (kept for canonicalization).", "typical_range_mM": [0.1, 5.0]}	1	2026-05-20 22:24:11.306852+01
597	chelator	DTPA	["diethylenetriaminepentaacetic acid", "pentetic acid"]	{"notes": "Stronger chelator than EDTA; not present in our 656 working-set corpus as of 2026-05-20 (kept for canonicalization).", "typical_range_mM": [0.1, 5.0]}	1	2026-05-20 22:24:11.306852+01
594	chelator	EDTA	["ethylenediaminetetraacetic acid", "edetic acid", "Na2EDTA", "disodium EDTA", "EDTA disodium", "ethylenediamine tetraacetate", "EDTA(Na2)", "Na4EDTA", "ethylenediaminetetracetic acid", "NaEDTA"]	{"metal_binding_logK_25C": {"Ca2+": 10.65, "Cu2+": 18.8, "Fe2+": 14.3, "Fe3+": 25.1, "Mg2+": 8.79, "Mn2+": 13.87, "Ni2+": 18.61, "Zn2+": 16.5, "source": "Martell-Smith CSC v.1 (NIST SRD 46)"}, "notes": "Most common chelator in NMR samples. At pH 7 dominantly HY^3- (~88%, net charge -3); contributes ~4.5 mM ionic strength per mM EDTA at pH 7. At 1+ mM essentially nullifies bulk divalent ions for genion.", "pKa": [{"source": "Martell-Smith CSC v.1", "step": 1, "value": 2.0}, {"source": "Martell-Smith CSC v.1", "step": 2, "value": 2.69}, {"source": "Martell-Smith CSC v.1", "step": 3, "value": 6.13}, {"source": "Martell-Smith CSC v.1", "step": 4, "value": 10.37}], "source": "Martell-Smith Critical Stability Constants v.1; NIST SRD 46", "species": [{"charge": 0, "label": "H4Y", "note": "Fully protonated; dominant pH < 2."}, {"charge": -1, "label": "H3Y-", "step": 1}, {"charge": -2, "label": "H2Y^2-", "step": 2}, {"charge": -3, "label": "HY^3-", "note": "Dominant species at pH 6.5-10; the form that chelates."}, {"charge": -4, "label": "Y^4-", "step": 4}], "typical_range_mM": [0.1, 5.0]}	1	2026-05-20 22:24:11.306852+01
595	chelator	EGTA	["ethylene glycol-bis(2-aminoethyl ether)-N,N,N',N'-tetraacetic acid", "EGTA acid"]	{"approx_net_charge_pH7": -2, "notes": "Selective for Ca2+ over Mg2+ (logK difference ~5.7 vs ~5.7 for EGTA/Ca vs EGTA/Mg). At pH 7 dominantly H2Y^2- (net charge -2). 2 proteins in working set.", "pKa": [{"source": "Martell-Smith CSC / NIST SRD 46", "step": 1, "value": 2.0}, {"source": "Martell-Smith CSC / NIST SRD 46", "step": 2, "value": 2.7}, {"source": "Martell-Smith CSC / NIST SRD 46", "step": 3, "value": 8.84}, {"source": "Martell-Smith CSC / NIST SRD 46", "step": 4, "value": 9.43}], "pKa_amine_1": 8.84, "pKa_amine_2": 9.43, "source": "Martell-Smith CSC v.1 (pKa); Tsien 1980 (Ca2+/Mg2+ selectivity)", "species": [{"charge": 0, "label": "H4Y", "step": null}, {"charge": -1, "label": "H3Y-", "step": 1}, {"charge": -2, "label": "H2Y^2-", "step": 2}, {"charge": -3, "label": "HY^3-", "step": 3}, {"charge": -4, "label": "Y^4-", "step": 4}], "typical_range_mM": [0.1, 5.0]}	1	2026-05-20 22:24:11.306852+01
599	chelator	NTA	["nitrilotriacetic acid", "nitrilotriacetate"]	{"notes": "Common in IMAC purification (Ni-NTA columns); not present in our 656 working-set corpus as of 2026-05-20 (kept for canonicalization)."} {"notes": "Citrate is both a buffer and a weak chelator. When recorded as a chelator, distinguish from buffer-role. Not currently extracted as chelator in our 656 working-set (citrate buffer entries handled by buffer dict)."} {"notes": "Common at 5-10% to dissolve hydrophobic ligands. Affects dielectric.", "typical_range_pct": [1, 10]}	1	2026-05-20 22:24:11.306852+01
598	chelator	citrate	["sodium citrate (chelator-role)", "citric acid (chelator-role)"]		1	2026-05-20 22:24:11.306852+01
606	cosolvent	DMSO	["dimethyl sulfoxide", "dimethylsulfoxide", "(CH3)2SO", "d-DMSO", "DMSO-d6", "deuterated DMSO"]	["notes": "Crystallization aid; sometimes left in NMR samples.", "typical_range_pct": [5, 30]}	1	2026-05-20 22:24:11.306852+01
611	cosolvent	MPD	["2-methyl-2,4-pentanediol", "methylpentanediol", "Hexylene glycol"]		1	2026-05-20 22:24:11.306852+01
612	cosolvent	PEG	["polyethylene glycol", "PEG-200", "PEG-400", "PEG-2000", "PEG-4000", "polyethyleneglycol"]		1	2026-05-20 22:24:11.306852+01
615	cosolvent	acetonitrile	["MeCN", "ACN", "CH3CN"]		1	2026-05-20 22:24:11.306852+01
608	cosolvent	ethanol	["EtOH", "ethyl alcohol", "C2H5OH"]		1	2026-05-20 22:24:11.306852+01
607	cosolvent	glycerol	["glycerin", "glycerine", "1,2,3-propanetriol", "d-glycerol", "deuterated glycerol"]		1	2026-05-20 22:24:11.306852+01
610	cosolvent	isopropanol	["2-propanol", "iPrOH", "propan-2-ol", "IPA"]		1	2026-05-20 22:24:11.306852+01
609	cosolvent	methanol	["MeOH", "methyl alcohol", "CH3OH"]		1	2026-05-20 22:24:11.306852+01
613	cosolvent	sucrose	["β-D-fructofuranosyl-α-D-glucopyranoside", "saccharose"]		1	2026-05-20 22:24:11.306852+01
614	cosolvent	trehalose	["α,α-trehalose", "α-D-glucopyranosyl-α-D-glucopyranoside"]		1	2026-05-20 22:24:11.306852+01
601	denaturant	GuHCl	["guanidine hydrochloride", "guanidinium hydrochloride", "guanidine HCl", "GdmHCl", "GdmHCl"]		1	2026-05-20 22:24:11.306852+01
605	denaturant	HFIP	["hexafluoroisopropanol", "hexafluoro-2-propanol", "HFIP"]		1	2026-05-20 22:24:11.306852+01
603	denaturant	SDS	["sodium dodecyl sulfate", "sodium dodecyl sulphate", "lauryl sulfate", "Na dodecyl sulfate"]		1	2026-05-20 22:24:11.306852+01
604	denaturant	TFE	["trifluoroethanol", "2,2,2-trifluoroethanol"]		1	2026-05-20 22:24:11.306852+01

ID	Class	Canonical name	Synonyms (JSON)	Parameters (JSON)	Version	Created
602	denaturant	guanidine thiocyanate	["guanidinium thiocyanate", "GuSCN"]	{"notes": "Stronger denaturant than GuHCl; rare in NMR samples."}	1	2026-05-20 22:24:11.306852+01
600	denaturant	urea	["carbamide", "carbonyl diamide"]	{"notes": "Above ~4 M denatures most proteins. Below ~2 M usually does not. Concentrations between are sample-dependent.", "typical_denaturing_range_M": [4.0, 8.0]}	1	2026-05-20 22:24:11.306852+01
622	detergent	CHAPS	["3-[(3-cholamidopropyl)dimethylammonio]-1-propanesulfonate"]	{"type": "zwitterionic"}	1	2026-05-20 22:24:11.306852+01
618	detergent	DDM	["n-dodecyl β-D-maltoside", "n-dodecyl-β-D-maltopyranoside", "dodecyl maltoside", "lauryl maltoside"]	{"notes": "Common membrane-protein solubilization detergent.", "type": "non-ionic"}	1	2026-05-20 22:24:11.306852+01
617	detergent	DHPC	["1,2-diheptanoyl-sn-glycero-3-phosphocholine", "diheptanoyl phosphatidylcholine", "DH(7)PC"]	{"notes": "Bicelle component; membrane-mimicking.", "type": "short-chain phospholipid"}	1	2026-05-20 22:24:11.306852+01
616	detergent	DPC	["dodecylphosphocholine", "n-dodecylphosphocholine", "FOS-cho-12", "Foscholine-12"]	{"notes": "Common micellar host for membrane-protein NMR. Presence indicates membrane protein context; standard solution-MD setup is not appropriate.", "type": "phosphocholine"}	1	2026-05-20 22:24:11.306852+01
620	detergent	LDAO	["lauryldimethylamine N-oxide", "N,N-dimethyldodecylamine N-oxide"]	{"type": "zwitterionic"}	1	2026-05-20 22:24:11.306852+01
621	detergent	LMNG	["lauryl maltose neopentyl glycol", "LMNG-100"]	{"type": "non-ionic"}	1	2026-05-20 22:24:11.306852+01
619	detergent	OG	["n-octyl β-D-glucoside", "octyl glucoside", "octyl-β-D-glucopyranoside"]	{"type": "non-ionic"}	1	2026-05-20 22:24:11.306852+01
623	detergent	Triton X-100	["Triton-X-100", "TX-100", "octylphenol ethoxylate", "polyethylene glycol p-(1,1,3,3-tetramethylbutyl)-phenyl ether"]	{"type": "non-ionic"}	1	2026-05-20 22:24:11.306852+01
542	ion	Br-	["Br", "Br(-)", "bromide", "bromide ion"]	{"charge": -1, "ff_parameter_source": "Joung-Cheatham 2008 (TIP3P)", "gm_x_genion_role": "nname", "gm_x_genion_token": "BR", "valence": 1}	1	2026-05-20 22:24:11.306852+01
534	ion	Ca2+	["Ca", "Ca++", "Ca(2+)", "Ca+2", "calcium", "calcium ion", "calcium(II)", "Ca(II)"]	{"charge": 2, "ff_parameter_source": "Mamatkulov-Schwierz 2018 (TIP3P); Li-Merz 12-6-4 for binding-site work", "gm_x_genion_role": "pname", "gm_x_genion_token": "CA", "valence": 2}	1	2026-05-20 22:24:11.306852+01
541	ion	Cl-	["Cl", "Cl(-)", "Cl-1", "chloride", "chloride ion"]	{"charge": -1, "ff_parameter_source": "Joung-Cheatham 2008 (TIP3P)", "gm_x_genion_role": "nname", "gm_x_genion_token": "CL", "valence": 1}	1	2026-05-20 22:24:11.306852+01
532	ion	Cs+	["Cs", "Cs(+)", "cesium", "caesium", "cesium ion"]	{"charge": 1, "ff_parameter_source": "Joung-Cheatham 2008 (TIP3P)", "gm_x_genion_role": "pname", "gm_x_genion_token": "CS", "valence": 1}	1	2026-05-20 22:24:11.306852+01
537	ion	Cu+	["Cu(I)", "Cu+1", "copper(I)", "cuprous"]	{"charge": 1, "ff_parameter_source": "Li-Merz 12-6-4 (bonded model)", "gm_x_genion_role": "pname", "gm_x_genion_token": "CU1", "valence": 1}	1	2026-05-20 22:24:11.306852+01
536	ion	Cu2+	["Cu", "Cu++", "Cu(2+)", "Cu+2", "copper", "copper ion", "copper(II)", "Cu(II)"]	{"charge": 2, "ff_parameter_source": "Li-Merz 12-6-4 (bonded model)", "gm_x_genion_role": "pname", "gm_x_genion_token": "CU", "valence": 2}	1	2026-05-20 22:24:11.306852+01
538	ion	Fe2+	["Fe", "Fe(II)", "Fe++", "Fe(2+)", "iron(II)", "ferrous"]	{"charge": 2, "ff_parameter_source": "Li-Merz 12-6-4 (bonded model)", "gm_x_genion_role": "pname", "gm_x_genion_token": "FE2", "valence": 2}	1	2026-05-20 22:24:11.306852+01
539	ion	Fe3+	["Fe(III)", "Fe(3+)", "Fe+3", "iron(III)", "ferric"]	{"charge": 3, "ff_parameter_source": "Li-Merz 12-6-4 (bonded model)", "gm_x_genion_role": "pname", "gm_x_genion_token": "FE", "valence": 3}	1	2026-05-20 22:24:11.306852+01
546	ion	H2PO4-	["H2P04", "H2P04(-)", "monobasic phosphate", "dihydrogen phosphate"]	{"charge": -1, "notes": "Dominant phosphate species at pH 5-7; treated as point charge for I.", "valence": 1}	1	2026-05-20 22:24:11.306852+01
545	ion	HPO4^2-	["HP04", "HP04(2-)", "HP04--", "monohydrogen phosphate", "dibasic phosphate"]	{"charge": -2, "notes": "Dominant phosphate species at pH 7-8; treated as point charge for I.", "valence": 2}	1	2026-05-20 22:24:11.306852+01
543	ion	I-	["I(-)", "iodide", "iodide ion"]	{"charge": -1, "ff_parameter_source": "Joung-Cheatham 2008 (TIP3P)", "gm_x_genion_role": "nname", "gm_x_genion_token": "IOD", "valence": 1}	1	2026-05-20 22:24:11.306852+01
529	ion	K+	["K", "K(+)", "K+1", "potassium", "potassium ion", "potassium cation"]	{"charge": 1, "ff_parameter_source": "Joung-Cheatham 2008 (TIP3P)", "gm_x_genion_role": "pname", "gm_x_genion_token": "K", "valence": 1}	1	2026-05-20 22:24:11.306852+01
531	ion	Li+	["Li", "Li(+)", "lithium", "lithium ion"]	{"charge": 1, "ff_parameter_source": "Joung-Cheatham 2008 (TIP3P)", "gm_x_genion_role": "pname", "gm_x_genion_token": "LI", "valence": 1}	1	2026-05-20 22:24:11.306852+01
533	ion	Mg2+	["Mg", "Mg++", "Mg(2+)", "Mg+2", "magnesium", "magnesium ion", "magnesium(II)", "Mg(II)"]	{"charge": 2, "ff_parameter_source": "Mamatkulov-Schwierz 2018 (TIP3P); Li-Merz 12-6-4 for binding-site work", "gm_x_genion_role": "pname", "gm_x_genion_token": "MG", "valence": 2}	1	2026-05-20 22:24:11.306852+01
540	ion	Mn2+	["Mn", "Mn(II)", "Mn++", "Mn(2+)", "manganese(II)"]	{"charge": 2, "ff_parameter_source": "Li-Merz 12-6-4 (bonded model)", "gm_x_genion_role": "pname", "gm_x_genion_token": "MN", "valence": 2}	1	2026-05-20 22:24:11.306852+01
530	ion	NH4+	["NH4", "NH4(+)", "NH+", "ammonium", "ammonium ion", "ammonium cation"]	{"charge": 1, "ff_parameter_source": "ff145B ions library", "gm_x_genion_role": "pname", "gm_x_genion_token": "NH4", "valence": 1}	1	2026-05-20 22:24:11.306852+01
528	ion	Na+	["Na", "Na(+)", "Na+1", "sodium", "sodium ion", "sodium cation", "sodium(I)"]	{"charge": 1, "ff_parameter_source": "Joung-Cheatham 2008 (TIP3P)", "gm_x_genion_role": "pname", "gm_x_genion_token": "NA", "valence": 1}	1	2026-05-20 22:24:11.306852+01
547	ion	PO4^3-	["P04", "P04(3-)", "P04---", "phosphate(3-)", "tribasic phosphate"]	{"charge": -3, "notes": "Dominant phosphate species at pH > 12; rare at biological pH.", "valence": 3}	1	2026-05-20 22:24:11.306852+01
544	ion	SO4^2-	["S04", "S04(2-)", "S04--", "sulfate", "sulphate", "sulfate ion"]	{"charge": -2, "notes": "Treated as point charge for I calculation; not used as a free counterion in genion.", "valence": 2}	1	2026-05-20 22:24:11.306852+01
535	ion	Zn2+	["Zn", "Zn++", "Zn(2+)", "Zn+2", "zinc", "zinc ion", "zinc(II)", "Zn(II)"]	{"charge": 2, "ff_parameter_source": "Li-Merz 12-6-4 (bonded model for coordination)", "gm_x_genion_role": "pname", "gm_x_genion_token": "ZN", "notes": "If coordinating CYS/HIS, requires bonded model, not bulk ion treatment.", "valence": 2}	1	2026-05-20 22:24:11.306852+01
548	ion	acetate-	["acetate", "OAc-", "CH3COO-", "AcO-", "ethanoate"]	{"charge": -1, "notes": "Buffer ion; see also buffer:acetate for pKa data.", "valence": 1}	1	2026-05-20 22:24:11.306852+01
640	isotopic_labeling	[U-13C]	["U-13C", "(U-13C)", "uniformly 13C labeled"]	{"atoms_enriched": ["C"], "uniform": true}	1	2026-05-20 22:24:11.306852+01
639	isotopic_labeling	[U-15N,13C,2H]	["U-15N,13C,2H", "U-15N/13C/2H", "triple labeled", "perdeuterated", "(U-15N,13C,2H)", "[U-(15)N,(13)C,(2)H]", "[U-15N,13C,2H]", "[U-2H; U-13C; U-15N]", "[U-2H, U-13C, U-15N]", "[U-2H;U-13C;U-15N]", "[U-(2)H; U-(13)C; U-(15)N]", "[2H; 13C; 15N]"]	{"atoms_enriched": ["N", "C", "H"], "notes": "Perdeuteration: backbone N and C uniformly labeled, side-chain H replaced with 2H. MD uses different H mass if explicit.", "uniform": true}	1	2026-05-20 22:24:11.306852+01

ID	Class	Canonical name	Synonyms (JSON)	Parameters (JSON)	Version	Created
638	isotopic_labeling	[U-15N,13C]	["U-15N,13C", "U-15N/13C", "(U-15N,13C)", "[U-(15)N, (13)C]", "uniformly 15N and 13C labeled", "double labeled", "[U-13C; U-15N]", "[U-13C, U-15N]", "[U-13C;U-15N]", "[U-15N, U-13C]", "[U-15N;U-13C]", "[U-(13)C; U-(15)N]"]	{"atoms_enriched": ["N", "C"], "uniform": true}	1	2026-05-20 22:24:11.306852+01
637	isotopic_labeling	[U-15N]	["U-15N", "uniform 15N", "(U-15N)", "[U-(15)N]", "uniformly 15N labeled", "15N uniform"]	{"atoms_enriched": ["N"], "notes": "Uniformly 15N-labeled; cheapest common labeling for backbone NMR.", "uniform": true}	1	2026-05-20 22:24:11.306852+01
641	isotopic_labeling	natural abundance	["NA", "unlabeled", "no labeling", "natural", "natural abundance only", ".".]	{"atoms_enriched": [], "notes": "Sample is not isotopically enriched.", "uniform": false}	1	2026-05-20 22:24:11.306852+01
632	metal_cofactor	Ca	["Ca(II)", "calcium cofactor", "EF-hand Ca", "CALCIUM (II) ION", "Calcium (II) Ion", "Ca(II) ion"]	{"common_coordinators": ["ASP/GLU carboxylate", "ASN/GLN amide O", "backbone O", "water"], "ff_parameter_source": "Mamatkulov-Schwierz 2018 bulk; Li-Merz 12-6-4 for binding site", "notes": "EF-hand binding sites have a specific coordination geometry.", "valence": 2}	1	2026-05-20 22:24:11.306852+01
628	metal_cofactor	Cu	["Cu(I)", "Cu(II)", "copper cofactor", "COPPER (I) ION", "COPPER (II) ION", "Copper (I) Ion", "Copper (II) Ion", "Cu(I) ion", "Cu(II) ion"]	{"common_coordinators": ["CYS Sy", "HIS Nε/Nδ", "MET Sδ"], "ff_parameter_source": "Li-Merz 12-6-4 (bonded)", "notes": "Redox-active. Plastocyanin, azurin, blue Cu proteins.", "valence_states": [1, 2]}	1	2026-05-20 22:24:11.306852+01
629	metal_cofactor	Fe	["Fe(II)", "Fe(III)", "iron cofactor", "Fe-S cluster", "heme iron", "iron(II/III)", "IRON (II) ION", "IRON (III) ION", "Iron (II) Ion", "Iron (III) Ion", "Fe(II) ion", "Fe(III) ion"]	{"common_coordinators": ["HIS Nε/Nδ", "CYS Sy", "porphyrin N", "ASP/GLU"], "ff_parameter_source": "Li-Merz 12-6-4 (bonded); heme requires custom rtp", "notes": "Heme is a distinct case – custom rtp/parameters needed.", "valence_states": [2, 3]}	1	2026-05-20 22:24:11.306852+01
631	metal_cofactor	Mg	["Mg(II)", "magnesium cofactor", "MAGNESIUM (II) ION", "Magnesium (II) Ion", "Mg(II) ion"]	{"common_coordinators": ["ASP/GLU carboxylate", "phosphate O", "water"], "ff_parameter_source": "Mamatkulov-Schwierz 2018 bulk; Li-Merz 12-6-4 for binding site", "notes": "Often as bulk ion; cofactor case requires the binding-site parameter set.", "valence": 2}	1	2026-05-20 22:24:11.306852+01
630	metal_cofactor	Mn	["Mn(II)", "manganese cofactor", "MANGANESE (II) ION", "Manganese (II) Ion", "Mn(II) ion"]	{"common_coordinators": ["ASP/GLU carboxylate", "HIS", "water"], "ff_parameter_source": "Li-Merz 12-6-4 (bonded)", "valence": 2}	1	2026-05-20 22:24:11.306852+01
627	metal_cofactor	Zn	["Zn(II)", "zinc cofactor", "Zn(2+) bound", "Zn(II) cofactor", "ZINC (II) ION", "Zinc (II) Ion", "Zn(II) ion", "zinc ion (II)"]	{"common_coordinators": ["CYS Sy", "HIS Nε/Nδ", "ASP/GLU carboxylate", "water"], "ff_parameter_source": "Li-Merz 12-6-4 (bonded)", "notes": "Zn fingers, hydrolases, dehydrogenases. CYS coordinating – force CYM.", "valence": 2}	1	2026-05-20 22:24:11.306852+01
635	ph_measurement_type	pD	["pD", "pD (corrected)", "p2H"]	{"correction": "Trust as given; no further correction.", "description": "Depositor claims a D-corrected pH for a D2O sample."}	1	2026-05-20 22:24:11.306852+01
636	ph_measurement_type	pD*	["pD*"]	{"correction": "Treat as pH* (apply Glasoe-Long when D2O > 50% AND pH > 8).", "description": "Ambiguous; some depositors use this for meter-reading in D2O without correction."}	1	2026-05-20 22:24:11.306852+01
633	ph_measurement_type	pH	["pH (aqueous)", "pH (uncorrected)"]	{"correction": "none", "description": "Normal aqueous pH-meter reading; H2O >= 95%."}	1	2026-05-20 22:24:11.306852+01
634	ph_measurement_type	pH*	["pH*", "pH (D2O)", "pH meter (D2O)", "uncorrected meter reading in D2O", "pHmeter", "pH apparent", "apparent pH", "meter reading"]	{"correction": "Glasoe-Long: pH_corrected = pH*_meter + 0.40 * D2O_fraction (NIST 2017 rule: only apply when D2O > 50% AND pH > 8).", "description": "Meter reading in a D2O-containing sample; the electrode reads the proton activity but the D2O fraction shifts it."}	1	2026-05-20 22:24:11.306852+01
626	preservative	PMSF	["phenylmethylsulfonyl fluoride", "phenylmethanesulfonyl fluoride"]	{"notes": "Serine protease inhibitor; sometimes used to prevent protein degradation during long acquisition."}	1	2026-05-20 22:24:11.306852+01
625	preservative	Thimerosal	["thiomersal", "ethyl(2-mercaptobenzoato-S)mercurate"]	{"notes": "Hg-containing preservative; rare in modern NMR samples."}	1	2026-05-20 22:24:11.306852+01
624	preservative	sodium azide	["NaN3", "Na N3", "azide", "Na azide"]	{"notes": "Microbial growth inhibitor; common at 0.02-0.05% (= ~3-8 mM). Very common in long NMR acquisitions. Toxic; not modeled in MD.", "typical_range_mM": [1.5, 8.0], "typical_range_pct": [0.01, 0.05]}	1	2026-05-20 22:24:11.306852+01
589	reducing_agent	BME	["β-mercaptoethanol", "beta-mercaptoethanol", "2-mercaptoethanol", "2-ME", "2ME", "ME", "mercaptoethanol", "d-BME", "d-2-mercaptoethanol", "d6-BME", "deuterated BME", "b-mercaptoethanol"]	{"effective_threshold_mM": 5.0, "notes": "Volatile, odorous. Required at higher concentration than DTT due to equilibrium with mixed disulfide.", "pKa": [{"source": "Whitesides 1977", "step": 1, "value": 9.6}], "pKa_thiol": 9.6, "source": "Whitesides, Lilburn, Szajewski 1977, J. Org. Chem. 42:332 (thiol pKa); Jocelyn 1972 (general thiol chemistry)", "species": [{"charge": 0, "label": "BME", "step": null}, {"charge": -1, "label": "BME-S-", "step": 1}], "typical_range_mM": [5.0, 50.0]}	1	2026-05-20 22:24:11.306852+01
588	reducing_agent	DTE	["dithioerythritol", "dte", "trans-2,3-dimercapto-1,4-butanediol"]	{"effective_threshold_mM": 1.0, "notes": "Diastereomer of DTT (erythro vs threo configuration). Same redox couple; slightly lower thiol pKa values. Interchangeable with DTT for protein-prep purposes.", "pKa": [{"source": "Cleland 1964", "step": 1, "value": 9.0}, {"source": "Cleland 1964", "step": 2, "value": 9.9}], "pKa_thiol_1": 9.0, "pKa_thiol_2": 9.9, "source": "Cleland 1964, Biochemistry 3:480", "species": [{"charge": 0, "label": "DTE", "step": null}, {"charge": -1, "label": "DTE-S-", "step": 1}, {"charge": -2, "label": "DTE-S^2-", "step": 2}], "typical_range_mM": [1.0, 10.0]}	1	2026-05-20 22:24:11.306852+01
587	reducing_agent	DTT	["dithiothreitol", "Cleland's reagent", "Cleland", "DTT (Cleland)", "trans-1,4-dimercapto-2,3-butanediol", "d-DTT", "d10-DTT", "d-dithiothreitol", "perdeuterated DTT", "d-DDT", "dtt-d10", "dithiothreitol", "dithiothreitol-D10"]	{"effective_threshold_mM": 1.0, "notes": "Active at pH near 7. Below pH 5 effectively inactive even at high concentration. Air-oxidation slow but real.", "pKa": [{"source": "Cleland 1964", "step": 1, "value": 9.2}, {"source": "Cleland 1964", "step": 2, "value": 10.1}], "pKa_thiol_1": 9.2, "pKa_thiol_2": 10.1, "source": "Cleland 1964, Biochemistry 3:480", "species": [{"charge": 0, "label": "DTT", "step": null}, {"charge": -1, "label": "DTT-S-", "step": 1}, {"charge": -2, "label": "DTT-S^2-", "step": 2}], "typical_range_mM": [1.0, 10.0]}	1	2026-05-20 22:24:11.306852+01
591	reducing_agent	GSH	["reduced glutathione", "glutathione (reduced)", "L-glutathione", "γ-L-glutamyl-L-cysteinylglycine"]	{"effective_threshold_mM": 1.0, "notes": "Plain 'glutathione' is ambiguous: GSH (reduced) vs GSSG (oxidized). Not present in our 656 working-set corpus as of 2026-05-20 (kept for canonicalization).", "typical_range_mM": [1.0, 10.0]}	1	2026-05-20 22:24:11.306852+01
590	reducing_agent	TCEP	["tris(2-carboxyethyl)phosphine", "TCEP-HCl", "TCEPHCl", "tris-carboxyethyl-phosphine"]	{"approx_net_charge_pH7": -2, "effective_threshold_mM": 0.5, "notes": "Active across wide pH range including acidic. Non-thiol mechanism (phosphine-phosphine oxide); doesn't react with iodoacetamide. Three carboxylate pKas in the 3-4.5 range plus phosphine pKa 7.66 give net charge approximately -2 at biological pH. 3 proteins in our 656 corpus.", "pKa_phosphine": 7.66, "source": "Burns, Butler, Moran, Whitesides 1991, J. Org. Chem. 56:2648 (phosphine pKa)", "typical_range_mM": [0.5, 5.0], "useful_pH_range": [1.5, 8.5]}	1	2026-05-20 22:24:11.306852+01
592	reducing_agent	free cysteine	["L-cysteine", "cysteine (free)", "Cys (free)", "free Cys", "L-Cys", "cysteine HCl"]	{"effective_threshold_mM": 1.0, "notes": "Distinguish from cysteine residues in the protein; component-vs-residue context. Not present in our 656 working-set corpus as of 2026-05-20 (kept for canonicalization).", "typical_range_mM": [1.0, 10.0]}	1	2026-05-20 22:24:11.306852+01

ID	Class	Canonical name	Synonyms (JSON)	Parameters (JSON)	Version	Created
593	reducing_agent	sulfite	["Na2SO3", "sodium sulfite", "Na sulfite"]	{"effective_threshold_mM": 1.0, "notes": "Not present in our 656 working-set corpus as of 2026-05-20 (kept for canonicalization).", "typical_range_mM": [1.0, 10.0]}	1	2026-05-20 22:24:11.306852+01
644	salt	CaCl	["CaCl"]	{"dissociates_to": [{"ion": "Ca2+", "n": 1}, {"ion": "Cl-", "n": 2}], "ionic_strength_per_mM": 3.0, "tentative": true, "why": "assumed CaCl2 (calcium chloride, the usual reagent); but the bmr7362 sample loop lists NaCl 5 mM, so it could be NaCl"}	1	2026-06-01 08:09:24.715501+01
553	salt	CaCl2	["calcium chloride", "Ca chloride", "Ca(Cl)2"]	{"dissociates_to": [{"ion": "Ca2+", "n": 1}, {"ion": "Cl-", "n": 2}], "ionic_strength_per_mM": 3.0}	1	2026-05-20 22:24:11.306852+01
645	salt	GaCl2	["GaCl2"]	{"dissociates_to": [{"ion": "Ga3+", "n": 1}, {"ion": "Cl-", "n": 2}], "ionic_strength_per_mM": 5.5, "tentative": true, "why": "assumed GaCl2 (gallium chloride) per the 1NI7 PDB metadata; could be a CaCl2 typo, and gallium is not a standard MD ion"}	1	2026-06-01 08:09:24.715501+01
558	salt	K2HPO4	["dipotassium phosphate", "K2 HP04", "potassium phosphate dibasic"]	{"dissociates_to": [{"ion": "K+", "n": 2}, {"ion": "HP04^2-", "n": 1}], "ionic_strength_per_mM": 3.0}	1	2026-05-20 22:24:11.306852+01
550	salt	KCl	["potassium chloride", "K chloride", "K+ Cl-", "potassium_chloride"]	{"dissociates_to": [{"ion": "K+", "n": 1}, {"ion": "Cl-", "n": 1}], "ionic_strength_per_mM": 1.0}	1	2026-05-20 22:24:11.306852+01
559	salt	KH2PO4	["monopotassium phosphate", "K H2P04", "potassium phosphate monobasic", "kipo4"]	{"dissociates_to": [{"ion": "K+", "n": 1}, {"ion": "H2P04-", "n": 1}], "ionic_strength_per_mM": 1.0}	1	2026-05-20 22:24:11.306852+01
642	salt	L-arginine	["L-ARG", "L-Arg", "arginine"]	{"approx_net_charge": 1, "ionic_strength_per_mM": 1.0, "notes": "charged amino-acid solubility additive (Arg/Glu pair, Hautbergue-Golovanov); net +1 below pH ~9; modeled as a 1:1 electrolyte for ionic strength", "source": "Thermo amino-acid pKa table; codex-proposed + verified 2026-06-01 (bmr7257)"}	1	2026-06-01 07:56:09.023798+01
643	salt	L-glutamate	["L-Glu", "L-glu", "glutamate", "L-glutamic acid"]	{"approx_net_charge": -1, "ionic_strength_per_mM": 1.0, "notes": "charged amino-acid solubility additive (Arg/Glu pair); net -1 above pH ~4.3; modeled as a 1:1 electrolyte for ionic strength", "source": "Thermo amino-acid pKa table; codex-proposed + verified 2026-06-01 (bmr7257)"}	1	2026-06-01 07:56:09.023798+01
552	salt	MgCl2	["magnesium chloride", "Mg chloride", "Mg(Cl)2"]	{"dissociates_to": [{"ion": "Mg2+", "n": 1}, {"ion": "Cl-", "n": 2}], "ionic_strength_per_mM": 3.0}	1	2026-05-20 22:24:11.306852+01
551	salt	NH4Cl	["ammonium chloride", "NH4 chloride"]	{"dissociates_to": [{"ion": "NH4+", "n": 1}, {"ion": "Cl-", "n": 1}], "ionic_strength_per_mM": 1.0}	1	2026-05-20 22:24:11.306852+01
563	salt	Na2EDTA	["disodium EDTA", "Na EDTA", "EDTA disodium", "EDTA(Na2)"]	{"dissociates_to": [{"ion": "Na+", "n": 2}, {"ion": "EDTA^2-", "n": 1}], "ionic_strength_per_mM": 3.0, "notes": "Chelator; categorize as 'chelator' not 'salt' for the chemistry pipeline."}	1	2026-05-20 22:24:11.306852+01
556	salt	Na2HPO4	["disodium phosphate", "Na2 HP04", "sodium phosphate dibasic", "dibasic sodium phosphate", "Na phosphate (dibasic)"]	{"dissociates_to": [{"ion": "Na+", "n": 2}, {"ion": "HP04^2-", "n": 1}], "ionic_strength_per_mM": 3.0, "notes": "Often paired with NaH2P04 to make a phosphate buffer."}	1	2026-05-20 22:24:11.306852+01
555	salt	Na2SO4	["sodium sulfate", "sodium sulphate", "Na sulfate", "(Na)2 SO4"]	{"dissociates_to": [{"ion": "Na+", "n": 2}, {"ion": "SO4^2-", "n": 1}], "ionic_strength_per_mM": 3.0}	1	2026-05-20 22:24:11.306852+01
549	salt	NaCl	["sodium chloride", "Na chloride", "Na+ Cl-", "salt", "table salt", "common salt"]	{"dissociates_to": [{"ion": "Na+", "n": 1}, {"ion": "Cl-", "n": 1}], "ionic_strength_per_mM": 1.0}	1	2026-05-20 22:24:11.306852+01
557	salt	NaH2PO4	["monosodium phosphate", "Na H2P04", "sodium phosphate monobasic", "monobasic sodium phosphate", "Na phosphate (monobasic)"]	{"dissociates_to": [{"ion": "Na+", "n": 1}, {"ion": "H2P04-", "n": 1}], "ionic_strength_per_mM": 1.0}	1	2026-05-20 22:24:11.306852+01
554	salt	ZnCl2	["zinc chloride", "Zn chloride", "Zn(Cl)2"]	{"dissociates_to": [{"ion": "Zn2+", "n": 1}, {"ion": "Cl-", "n": 2}], "ionic_strength_per_mM": 3.0}	1	2026-05-20 22:24:11.306852+01
561	salt	ammonium acetate	["NH4OAc", "NH4 acetate", "AmOAc"]	{"dissociates_to": [{"ion": "NH4+", "n": 1}, {"ion": "acetate-", "n": 1}], "ionic_strength_per_mM": 1.0}	1	2026-05-20 22:24:11.306852+01
560	salt	ammonium sulfate	["(NH4)2SO4", "NH4 sulfate", "ammonium sulphate", "AmSO4"]	{"dissociates_to": [{"ion": "NH4+", "n": 2}, {"ion": "SO4^2-", "n": 1}], "ionic_strength_per_mM": 3.0, "notes": "Common precipitant; some NMR samples have 0.5-1 M of this."}	1	2026-05-20 22:24:11.306852+01
562	salt	sodium acetate	["NaOAc", "Na acetate", "Na(CH3COO)", "AcONa", "d-NaOAc", "d-acetate", "d3-acetate", "d4-acetate", "deuterated sodium acetate", "NaAc", "sodium acetate D3"]	{"dissociates_to": [{"ion": "Na+", "n": 1}, {"ion": "acetate-", "n": 1}], "ionic_strength_per_mM": 1.0, "notes": "Acetate at pH < 4 is partially protonated (acetic acid pKa 4.76); at pH > 5 essentially fully ionized."}	1	2026-05-20 22:24:11.306852+01